Median: I

for Ungrouped data:

When n is odd, Median $= \left(\frac{n+1}{2}\right)^{th}$ observation

When n is even, Median $= \frac{\frac{n}{2}+\left(\frac{n}{2}+1\right)}{2}$ th observation

for grouped data:

$$\text{Median} = L + \frac{\frac{n}{2} - F_c}{f_m} \times c$$

where,

L = Lower limit of the median class

n = Total number of observation

F_c = Cumulative frequency of the class just proceeding the median class

f_m = Frequency of the median class

c = length of the median class

P.T.O

Mode:

For ungroped data:

Mode = Maximum frequency

For grouped data:

$$Mode = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times c$$

where,

L = Lower limit of the modal class

Δ_1 = Difference between frequency of the modal class and pre modal class

Δ_2 = " " " " " " and post modal class

c = length of the modal class

Quartiles:

For grouped data;

$$Quartiles, Q_i = L + \frac{\frac{i \times n}{4} - F_c}{f_q} \times c$$

where, c = The size of class interval

L = lower limit of quartile class

F_c = Proceeding cumulative frequency of Quartile class

f_q = frequency of the Quartile class.

Deciles:

For grouped data;

$$\text{deciles}, D_i = L + \frac{\frac{i \times n}{10} - F_c}{f_D} \times c$$

Percentiles:

For grouped data!

$$\text{Percentiles}, P_i = L + \frac{\frac{i n}{100} - F_c}{f_p} \times C$$

4. The following data shows the yearly temperature of north Bengal Bangladesh:

Temperature (°C)	5–10	10–15	15–20	20–25	25–30	30–35
of days	2	4	11	7	4	1

i) Find average yearly temperature ii) Compute the modal temperature iii) Using the cumulative frequency curve, determine the median temperature.

P.T.O

=>

Temperature (°C)	f_i	x_i	$f_i x_i$	F_c
5 − 10	2	7.5	15	2
10 − 15	4	12.5	50	6
15 − 20	11	17.5	192.5	17
20 − 25	7	22.5	157.5	24
25 − 30	4	27.5	110	28
30 − 35	1	32.5	32.5	29
	$\sum f_i = 29$		$\sum f_i x_i = 557.5$	

i) Average yearly temperature, $\overline{X} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i}$

$$= \frac{557.5}{29}$$

$$= 19.22 \text{ // (Ans)}$$

ii) Modal temperature, $= L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times C$

$$= 15 + \frac{(11-4)}{(11-4)+(11-7)} \times 5$$

$$= 15 + \frac{7}{7+4} \times 5$$

$$= 15 + \frac{35}{11}$$

$$= \frac{165+35}{11} = \frac{200}{11}$$

$$= 18.18 \text{ // (Ans)}$$

$$\therefore \cancel{n/2} = \frac{29}{2} = 15.5$$

ii) The median temperature $= L + \frac{n/2 - F_c}{f_m} \times C$

$$= 15 + \frac{15.5 - 6}{11} \times 5$$

$$= 19.31 \text{ '' (Ans)}$$

$= \cancel{15+}$

4. The frequency distribution given below refers to the blood cholesterol levels (given in milligrams per deciliter) of 50 patient admitted into the chittagong medical college hospital,

Cholesterol levels	149-169	169-189	189-209	209-229	229-249	249 to 269
No. of patient	4	6	15	13	7	5

i) Compute Mode and Median

ii) locate median graphically

iii) Locate mode graphically

iv) Compute Q_3, D_7 and P_{78}

P. T. O

⇒

Class interval	f_i	x_i	$f_i x_i$	F_c
149–169	4	159	636	4
169–189	6	179	1074	10
189–209	15	199	2985	25
209–229	13	219	2847	38
229–249	7	239	1673	45
249–269	5	249	1245	50
	$n=50$		$\sum f_i x_i = 10460$	

i) Mode $= L + \frac{\Delta_1}{\Delta_1+\Delta_2} \times c$

$\Delta_1 = 15-6 = 9$

$\Delta_2 = 15-13 = 2$

$= 189 + \frac{9}{9+2} \times 20$

$= 205.36$ (Ans)

$\therefore n/2 = 50/2 = 25$

Median $= L + \frac{n/2 - F_c}{f_m} \times c$

$= 189 + \frac{25 - 10}{15} \times 20$

$= 189 + \frac{15}{15} \times 20$

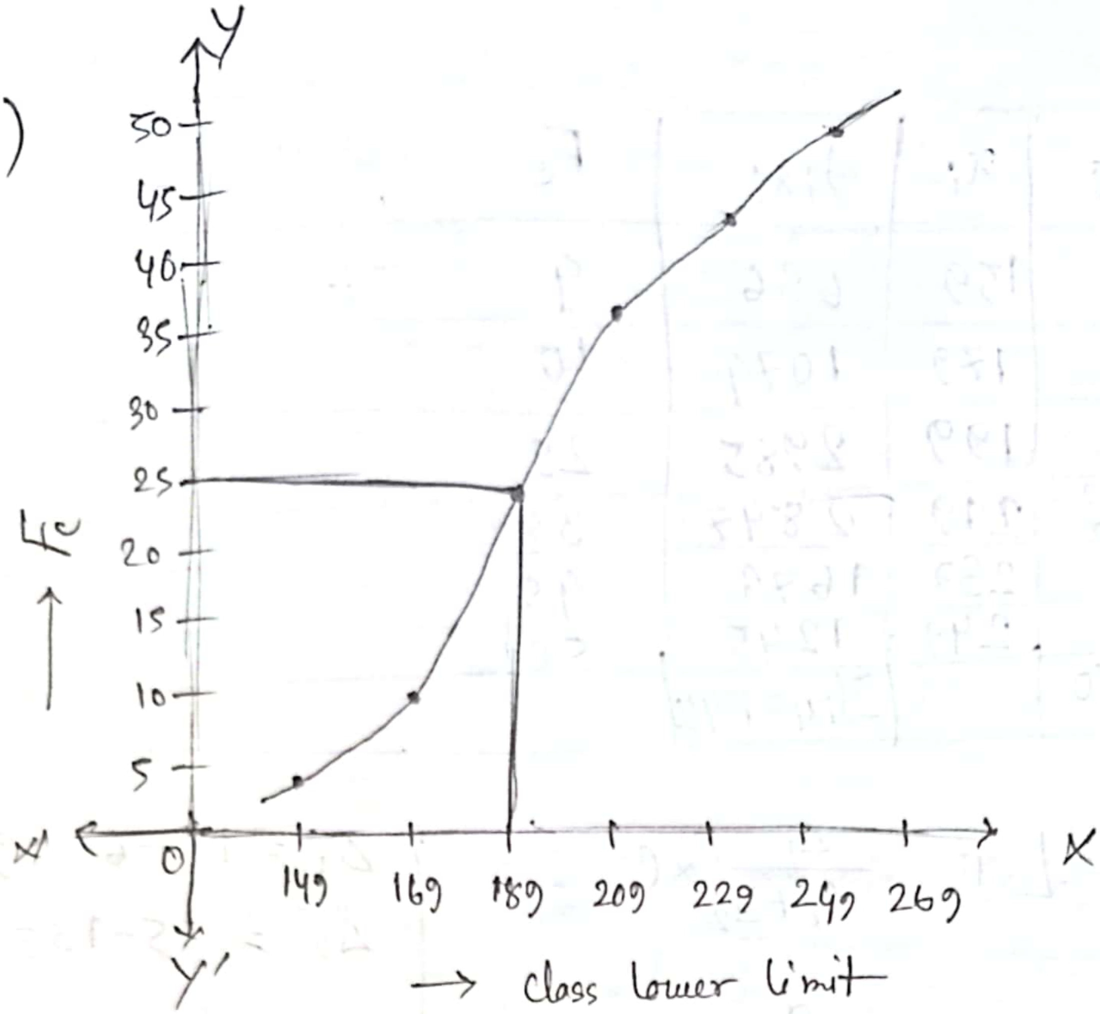
$= 209$ (Ans)

$= 189 + \frac{25-25}{15} \times 20$

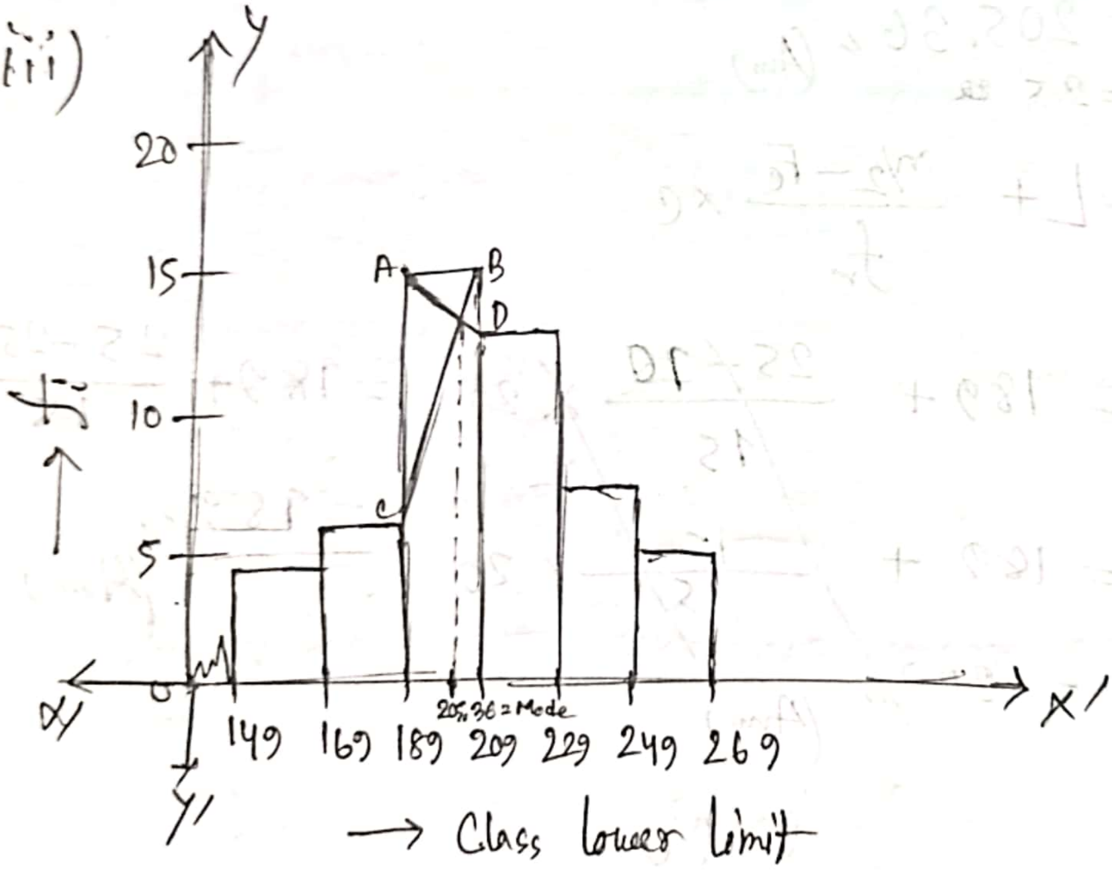
$= 189$ (Ans)

P.T.O

ii)


Y
50
45
40
35
30
25
20
15
10
5
fc
X'
0
149
169
189
209
229
249
269
X
Y'
Class lower limit

iii)


Y
20
15
10
5
fi
A
B
D
C
X'
0
205.36 = Mode
149
169
189
209
229
249
269
X'
Y'
Class lower limit

iv) $Q_3 = L + \dfrac{\frac{i \times n}{4} - F_c}{f_q} \times c$ $\left[\dfrac{i \times n}{4} = \dfrac{3 \times 50}{4} = 37.5 \right.$

$= 209 + \dfrac{37.5 - 25}{13} \times 20$

$= 228.23$ (Ans)

$D_7 = L + \dfrac{\frac{i \times n}{10} - F_c}{f_D} \times c$ $\left[\dfrac{i \times n}{10} = \dfrac{7 \times 50}{10} = 35 \right.$

$= 209 + \dfrac{35 - 25}{13} \times 20$

$= 224.38$ (Ans)

$P_{78} = L + \dfrac{\frac{i \times n}{100} - F_c}{f_p} \times c$ $\left[\dfrac{i \times n}{100} = \dfrac{78 \times 50}{100} = 39 \right.$

$= 229 + \dfrac{39 - 38}{7} \times 20$

$= 231.85$ (Ans)

P. T. O

Measures of central tendecy: A measure of central tendecy is a typical value around which others figure congregate.

Measures of Central Tendency

- Mean
  - Arithmetic Mean
  - Geometric Mean
  - Harmonic Mean
- Median
- Mode

Q. Describe the method of determining geometric mean and harmonic mean.

⇒ The method of geometric mean are given below:

1) Multiply all values together to get their product.

2) Find the n^{th} root of the product (n is the number of values).

The method of harmonic mean are given below:

1) Calculate the reciprocal of each value ($1/x_1, 1/x_2$ ---

2) Find the average of reciprocals obtained from Step 1

3) Finally take the reciprocal of the average obtained in Step 2

4). Explain clearly the concept of arithmetic mean, geometric mean and harmonic mean. Point out some of the applications for these concepts.

→ **Arithmetic Mean:** Arithmetic mean is the simplest measure of central tendency and is the ratio of the sum of the items to the number of items. It is denoted by $\bar{x}$.

Application or Uses of A.M:

① To analyze economics data, inflation rate and income distribution

② To describe patient data.

③ To analyze weather and climate data.

Geometric Mean: The geometric mean of a set of n non ~~zeo~~ zero positive observation's is the ~~the~~ n^{th} root of their product.

Application or Uses of G.M:

— ① To find the rate of population growth & the rate of interest

② In the construction of index number

Harmonic Mean: The Harmonic mean of a set of n non-zero observations $x_1, x_2, \ldots x_n$ in a series is the reciprocal of the arithmetic mean of the reciprocal.

Applications or Uses of H.M:

① It is used for calculating average speed of automobiles.

P. T. O

Q. Under what circumstances is it not possible to calculate the geometric mean?

⇒ These condition is not possible to calculate the geometric mean is given below;

① Negative Values
② Zero Values
③ Complex Numbers
④ Missing Data

Q. For two non-zero positive observation show that,
$A.M \geq G.M \geq H.M$

⇒ Let x_1 and x_2 be two non-zero positive observation then,

$$A.M = \frac{x_1+x_2}{2}$$

$$G.M = (x_1 \cdot x_2)^{1/2} = \sqrt{x_1 \cdot x_2}$$

$$H.M = \frac{2}{1/x_1 + 1/x_2}$$

Then,

$$\left(\sqrt{x_1} - \sqrt{x_2}\right)^2 \geq 0$$

$$\text{or, } (\sqrt{x_1})^2 - 2\sqrt{x_1} \cdot \sqrt{x_2} + (\sqrt{x_2})^2 \geq 0$$

$$\text{or, } x_1 - 2\sqrt{x_1 . x_2} + x_2 \geqslant 0$$

$$\text{or, } x_1 + x_2 \geqslant 2\sqrt{x_1 x_2}$$

$$\text{or, } \frac{x_1 + x_2}{2} \geqslant \sqrt{x_1 x_2}$$

$$\text{or, } A.M \geqslant G.M \quad \text{——} (1)$$

And Now,

$$\left(\frac{1}{\sqrt{x_1}} - \frac{1}{\sqrt{x_2}}\right)^2 \geqslant 0$$

$$\text{or, } \left(\frac{1}{\sqrt{x_1}}\right)^2 - 2\frac{1}{\sqrt{x_1}} . \frac{1}{\sqrt{x_2}} + \left(\frac{1}{\sqrt{x_2}}\right)^2 \geqslant 0$$

$$\text{or, } \frac{1}{x_1} - 2\frac{1}{\sqrt{x_1 . x_2}} + \frac{1}{x_2} \geqslant 0$$

$$\text{or, } \frac{1}{x_1} + \frac{1}{x_2} \geqslant \frac{2}{\sqrt{x_1 x_2}}$$

$$\text{or, } \sqrt{x_1 . x_2} \geqslant \frac{2}{1/x_1 + 1/x_2}$$

$$\text{or, } G.M \geqslant H.M \quad \text{——} (2)$$

From eq^n (1) & (2) $\Rightarrow$

$$A.M \geqslant G.M \geqslant H.M$$ (Showed) (Am)